Computer Communication Networks

Course Code: CSE248 | Spring 2026

GNS3 Network Simulation Project

Project Report

Project Title: SOHO Network Design & GNS3 Simulation

Network Type: *Small Office / Home Office (SOHO)*

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Demo Video Link: https://drive.google.com/file/d/1-cCwXjRjDSv9fVDhJSJQ5kgovsitOLLv/view

GitHub Repository: https://github.com/eyk-1/SOHO-network-gns3

1. Summary

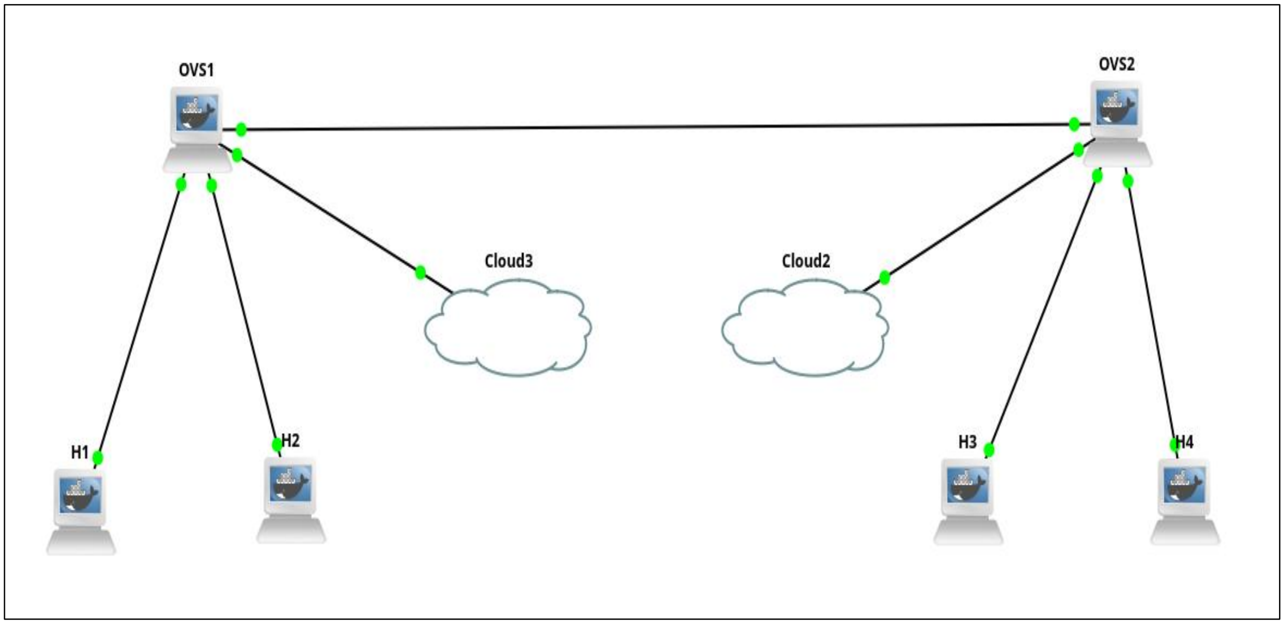
This report documents the design, configuration, and verification of a Small Office / Home Office (SOHO) network built and simulated using GNS3 on an Ubuntu 24.04 virtual machine hosted in VirtualBox. The project was completed individually as part of the CSE248 Computer Communication Networks course.

The network consists of one Cisco C3725 router, one Layer 2 Ethernet switch, and four VPCS end-user workstations. The implemented services include DHCP for automatic IP assignment, NAT/PAT for simulated internet access, a static default route, and an extended ACL acting as a basic firewall on the WAN interface. All devices were successfully configured and full connectivity was verified through ping tests and router show commands.

2. Network Design

2.1 Topology Overview

The SOHO network follows a simple star topology with the router as the central gateway. The switch connects all four PCs on the LAN side, while the router's WAN interface faces the simulated ISP. The design meets all minimum requirements for the SOHO network type as specified in the project brief.



2.2 Device List

Device	Hostname	Model	Role
Router	**R1**	**Cisco C3725**	*Default gateway, DHCP server, NAT, ACL*
Switch	**SW1**	**GNS3 Ethernet Switch**	*Layer 2 switching for LAN devices*
PC1	**PC1**	**GNS3 VPCS**	*End-user workstation*
PC2	**PC2**	**GNS3 VPCS**	*End-user workstation*
PC3	**PC3**	**GNS3 VPCS**	*End-user workstation*
PC4	**PC4**	**GNS3 VPCS**	*End-user workstation*

3. IP Addressing Table

3.1 Subnet Design

Segment	Network Address	Prefix	Usable Range	Purpose
LAN	192.168.1.0	**/24**	**192.168.1.1 – .254**	*Internal SOHO LAN*
WAN	203.0.113.0	**/30**	**203.0.113.1 – .2**	*Router to ISP link*

3.2 Device IP Address Table

Device	Interface	IP Address	Subnet Mask	Gateway	DHCP?
R1	**fa0/0 (LAN)**	192.168.1.1	255.255.255.0	—	**No**
R1	**fa0/1 (WAN)**	203.0.113.2	255.255.255.252	203.0.113.1	**No**
SW1	**VLAN 1 (Mgmt)**	192.168.1.2	255.255.255.0	192.168.1.1	**No**
PC1	**eth0**	192.168.1.10	255.255.255.0	192.168.1.1	**Yes**
PC2	**eth0**	192.168.1.11	255.255.255.0	192.168.1.1	**Yes**
PC3	**eth0**	192.168.1.12	255.255.255.0	192.168.1.1	**Yes**
PC4	**eth0**	192.168.1.13	255.255.255.0	192.168.1.1	**Yes**

4. Device Configurations

4.1 Router R1 — show ip interface brief

```
R1#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
FastEthernet0/0            192.168.1.1     YES NVRAM  up                    up
FastEthernet0/1            203.0.113.2     YES manual up                    up
NVI0                       192.168.1.1     YES unset  up                    up
```

4.2 Router R1 — show running-config

```
R1#show running-config
Building configuration...

Current configuration : 1477 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
!
hostname R1
!
no ip dhcp use vrf connected
ip dhcp excluded-address 192.168.1.1 192.168.1.9
!
ip dhcp pool SOHO_POOL
   network 192.168.1.0 255.255.255.0
   default-router 192.168.1.1
   dns-server 8.8.8.8
!
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 ip nat inside
 ip virtual-reassembly
 duplex auto
 speed auto
!
interface FastEthernet0/1
 ip address 203.0.113.2 255.255.255.252
 ip access-group FIREWALL in
 duplex auto
 speed auto
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip nat inside source list 1 interface FastEthernet0/1 overload
!
```

```
ip access-list extended FIREWALL
 permit tcp 192.168.1.0 0.0.0.255 any established
 permit icmp 192.168.1.0 0.0.0.255 any
 deny   ip any any
!
access-list 1 permit 192.168.1.0 0.0.0.255
!
end
```

5. Routing Protocol Analysis

5.1 Routing Method

Static routing was used for this SOHO network. A single default route (0.0.0.0/0) was configured on R1 pointing to the simulated ISP gateway at 203.0.113.1. This is appropriate for a SOHO network as there is only one exit point and dynamic routing protocols such as OSPF or RIP would be unnecessary overhead.

5.2 Routing Table — show ip route

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 203.0.113.1 to network 0.0.0.0

     203.0.113.0/30 is subnetted, 1 subnets
C       203.0.113.0 is directly connected, FastEthernet0/1
C    192.168.1.0/24 is directly connected, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 203.0.113.1
```

The routing table shows two directly connected networks (LAN and WAN) and one static default route marked with S*, confirming the gateway of last resort is correctly set to 203.0.113.1.

6. Services Configuration

6.1 DHCP

DHCP was configured on R1 to automatically assign IP addresses to all four PCs. Addresses 192.168.1.1 through 192.168.1.9 were excluded for infrastructure use. The DHCP pool assigns addresses from 192.168.1.10 onwards with the default gateway set to 192.168.1.1 and DNS set to 8.8.8.8.

6.2 DHCP Lease Results

```
PC1> dhcp
DDORA IP 192.168.1.10/24 GW 192.168.1.1

PC2> dhcp
DDORA IP 192.168.1.11/24 GW 192.168.1.1

PC3> dhcp
DDORA IP 192.168.1.12/24 GW 192.168.1.1

PC4> dhcp
DDORA IP 192.168.1.13/24 GW 192.168.1.1
```

6.3 DHCP Binding Table

```
R1#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address          Client-ID/              Lease expiration        Type
                    Hardware address/
                    User name
192.168.1.10        0100.5079.6668.00       Mar 02 2002 12:06 AM    Automatic
192.168.1.11        0100.5079.6668.01       Mar 02 2002 12:07 AM    Automatic
192.168.1.12        0100.5079.6668.02       Mar 02 2002 12:07 AM    Automatic
192.168.1.13        0100.5079.6668.03       Mar 02 2002 12:07 AM    Automatic
```

6.4 NAT / PAT

Network Address Translation with Port Address Translation (NAT overload) was configured on R1. The inside interface is fa0/0 (LAN) and the outside interface is fa0/1 (WAN). ACL 1 permits the 192.168.1.0/24 subnet for translation, allowing all four PCs to share the single public WAN IP address 203.0.113.2 for outbound internet traffic.

6.5 Firewall ACL

An extended ACL named FIREWALL was applied inbound on fa0/1 (WAN interface) to act as a basic firewall. It permits TCP return traffic for established sessions initiated from the LAN, permits ICMP from the LAN, and denies all other inbound IP traffic from the WAN.

7. Testing & Verification

7.1 PC to Gateway Ping Test

```
PC1> ping 192.168.1.1
84 bytes from 192.168.1.1 icmp_seq=1 ttl=255 time=10.164 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=255 time=3.012 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=255 time=7.208 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=255 time=6.221 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=255 time=3.185 ms
```

7.2 PC to PC Ping Test (PC1 to PC4)

```
PC1> ping 192.168.1.13
84 bytes from 192.168.1.13 icmp_seq=1 ttl=64 time=0.180 ms
84 bytes from 192.168.1.13 icmp_seq=2 ttl=64 time=0.266 ms
84 bytes from 192.168.1.13 icmp_seq=3 ttl=64 time=0.256 ms
84 bytes from 192.168.1.13 icmp_seq=4 ttl=64 time=0.285 ms
84 bytes from 192.168.1.13 icmp_seq=5 ttl=64 time=0.197 ms
```

7.3 Verification Summary

Test	Result	Status
PC1 ping to R1 (192.168.1.1)	***5/5 packets, avg 6ms***	**PASS**
PC1 ping to PC4 (192.168.1.13)	***5/5 packets, avg 0.2ms***	**PASS**
DHCP lease — PC1	***192.168.1.10 assigned***	**PASS**
DHCP lease — PC2	***192.168.1.11 assigned***	**PASS**
DHCP lease — PC3	***192.168.1.12 assigned***	**PASS**
DHCP lease — PC4	***192.168.1.13 assigned***	**PASS**
Default route configured	***S* 0.0.0.0/0 via 203.0.113.1***	**PASS**
NAT overload configured	***ACL 1 + fa0/1 overload***	**PASS**
Firewall ACL applied	***FIREWALL inbound on fa0/1***	**PASS**

8. Challenges Faced

8.1 GNS3 Server Not Starting

When GNS3 was first launched, it returned a 'Connection refused (localhost:3080)' error. The issue was that the GNS3 server process was not starting automatically. This was resolved by manually launching the server from the terminal using the 'gns3server' command before opening the GNS3 GUI.

8.2 Memory Allocation Warning

During the show running-config command, a MALLOCFAIL memory allocation error appeared. This indicated the router VM was running low on RAM. The issue was resolved by stopping R1, increasing the allocated RAM from 128MB to 256MB in the GNS3 device settings, and restarting the router.

8.3 Default Route Not Persisting

The static default route was not appearing in the routing table after initial configuration. Investigation revealed that fa0/1 (WAN interface) was still administratively down, causing the route to be rejected. The issue was fixed by assigning an IP address to fa0/1 and bringing it up with 'no shutdown' before re-entering the static route.

8.4 Ubuntu Safe Graphics Mode Required

During the initial Ubuntu installation in VirtualBox, the standard boot option produced a vmwgfx graphics driver error and a black screen. The issue was resolved by selecting the 'Ubuntu (safe graphics)' option from the GRUB boot menu, which bypassed the incompatible graphics driver and allowed the installer to load successfully.

9. Division of Work

This project was completed individually. All design, configuration, testing, and documentation tasks were performed by the student listed on the title page.

Week	Phase	Tasks Completed	Responsible
Week 1	**Setup & Proposal**	*Installed VirtualBox, Ubuntu 24.04, GNS3. Imported Cisco C3725 IOS image. Built 2-router test topology. Submitted Phase 1 proposal report.*	**Emad Yar Khan**
Week 2	**Design & Build**	*Built full SOHO topology in GNS3. Added switch and 4 VPCS. Assigned IPs to all interfaces. Configured DHCP pool.*	
Week 3	**Testing & Documentation**	*Configured NAT/PAT and extended ACL firewall. Brought up WAN interface. Set default route. Verified all connectivity. Captured all show command outputs.*	
Week 4	**Final Submission**	*Wrote final report. Recorded demo video. Exported VM as .ova file. Uploaded all deliverables.*	

10. Conclusion

This project successfully demonstrated the design and simulation of a functional SOHO network using GNS3. All required components were implemented including DHCP, NAT/PAT, static routing, and a basic ACL firewall. Full connectivity was verified between all devices on the network.

The project provided hands-on experience with Cisco IOS configuration, IP addressing and subnetting, and core network services that are directly applicable to real-world SOHO deployments. Key lessons learned include the importance of bringing interfaces up before applying routes, proper RAM allocation for Dynamips-based routers, and methodical troubleshooting when connectivity issues arise.

Possible improvements for future iterations include implementing a DNS server locally within the GNS3 topology, adding a second router to simulate a more realistic ISP connection, and exploring dynamic routing protocols such as OSPF for a more scalable design.

11. References

1. Cisco Systems, Inc. (2024). Cisco IOS IP Addressing Services Configuration Guide. Available at: https://www.cisco.com/c/en/us/tech/ip/ip-addressing/index.html
2. GNS3 Technologies Inc. (2024). GNS3 Documentation - Getting Started. Available at: https://docs.gns3.com/docs/
3. Hegde, P. (2024). Cisco Images for GNS3 and EVE-NG [GitHub Repository]. GitHub. Available at: https://github.com/hegdepavankumar/Cisco-Images-for-GNS3-and-EVE-NG